

```
1 #include <stdio.h>
2 #include <string.h>
3
4 int main()
5 {
6     int marks[5] = {80, 75, 90, 85, 95};
7
8     printf("Student Marks\n");
9
10    for(int i=0; i<=5; i++)
11        // Error: Loop accesses marks[5], which is out of bounds for an array of size 5.
12        // Solution: for(int i=0; i<5; i++)
13    {
14        printf("%d ", marks[i]);
15    }
16
17    printf("\n");
18
19    int total = 0;
20
21    for(int i=0; i<5; i++)
22    {
23        total += marks[i];
24    }
25
26    float average;
27
28    average = total / 5;
```

```
29 // Warning: Integer division occurs before assignment to float.
30 // Solution: average = (float)total / 5;
31
32 printf("Average = %.2f\n", average);
33
34 printf("\n");
35
36 int numbers[10];
37
38 for(int i=1; i<=10; i++)
39 // Error: Array index 10 is out of bounds and index 0 is never initialized.
40 // Solution: for(int i=0; i<10; i++)
41 {
42     numbers[i] = i * 10;
43 }
44
45 printf("Numbers Array\n");
46
47 for(int i=0; i<10; i++)
48 {
49     printf("%d ", numbers[i]);
50 }
51
52 printf("\n");
53
54 char name[10];
55
56 strcpy(name, "ComputerScience");
```

```
57 // Error: Buffer overflow. Destination array size is too small.
58 // Solution: char name[20]; strcpy(name, "ComputerScience");
59
60 printf("%s\n", name);
61
62 printf("\n");
63
64 char city[20] = "Jaipur";
65
66 city = "Delhi";
67 // Error: Cannot assign a string literal directly to a character array.
68 // Solution: strcpy(city, "Delhi");
69
70 printf("%s\n", city);
71
72 printf("\n");
73
74 char str1[20] = "Hello";
75 char str2[20] = "Hello";
76
77 if(str1 == str2)
78 // Error: Comparing string addresses instead of string contents.
79 // Solution: if(strcmp(str1, str2) == 0)
80 {
81     printf("Strings are Equal\n");
82 }
83 else
84 {
```

```
85     printf("Strings are Different\n");
86 }
87
88 printf("\n");
89
90 char password[8];
91
92 printf("Enter Password: ");
93 gets(password);
94 // Error: gets() is unsafe and may cause buffer overflow.
95 // Solution: fgets(password, sizeof(password), stdin);
96
97 printf("Password = %s\n", password);
98
99 printf("\n");
100
101 char firstName[20] = "Robil";
102 char lastName[20] = "Varshney";
103
104 strcat(firstName, lastName);
105 // Warning: Possible buffer overflow if combined string exceeds array size.
106 // Solution: Use a larger buffer or strncat() with bounds checking.
107
108 printf("%s\n", firstName);
109
110 printf("\n");
111
112 char grade[5];
```

```
113
114  grade[0] = 'A';
115  grade[1] = '+';
116
117  printf("%s\n", grade);
118  // Error: String is not null-terminated before printing with %s.
119  // Solution: grade[2] = '\0';
120
121  printf("\n");
122
123  int arr[5] = {1,2,3,4,5};
124
125  printf("%d\n", arr[-1]);
126  // Error: Negative array index causes undefined behavior.
127  // Solution: Access only valid indices 0 to 4.
128
129  printf("\n");
130
131  char department[10];
132
133  scanf("%s", department);
134  // Warning: No width limit specified; may cause buffer overflow.
135  // Solution: scanf("%9s", department);
136
137  printf("%s\n", department);
138
139  printf("\n");
140
```

```
141 char email[15] = "student@gmail.com";
142 // Error: Array size is too small for the string including null terminator.
143 // Solution: char email[18] = "student@gmail.com";
144
145 printf("%s\n", email);
146
147 printf("\n");
148
149 int scores[3];
150
151 scores[0] = 90;
152 scores[1] = 80;
153 scores[2] = 70;
154 scores[3] = 60;
155 // Error: Index 3 is out of bounds for an array of size 3.
156 // Solution: Declare scores[4] or remove this assignment.
157
158 printf("%d\n", scores[3]);
159 // Error: Accessing out-of-bounds array element.
160 // Solution: Access only indices 0 to 2.
161
162 printf("\n");
163
164 char subject[10] = "Programming";
165 // Error: Array size is too small for the string including null terminator.
166 // Solution: char subject[12] = "Programming";
167
168 printf("%s\n", subject);
```

```
169
170  printf("\n");
171
172  char username[20];
173
174  printf("Enter Username: ");
175  scanf("%s", username);
176  // Warning: No width limit specified; may cause buffer overflow.
177  // Solution: scanf("%19s", username);
178
179  if(username == "admin")
180  // Error: Comparing string addresses instead of contents.
181  // Solution: if(strcmp(username, "admin") == 0)
182  {
183      printf("Valid User\n");
184  }
185  else
186  {
187      printf("Invalid User\n");
188  }
189
190  printf("\n");
191
192  char mobile[10];
193
194  strcpy(mobile, "987654321012");
195  // Error: Buffer overflow. String exceeds destination array size.
196  // Solution: char mobile[13]; strcpy(mobile, "987654321012");
```

```
197
198  printf("%s\n", mobile);
199
200  printf("\n");
201
202  int attendance[5] = {1,1,1,1,1};
203
204  int present = 0;
205
206  for(int i=0; i<=5; i++)
207  // Error: Loop accesses attendance[5], which is out of bounds.
208  // Solution: for(int i=0; i<5; i++)
209  {
210      present += attendance[i];
211  }
212
213  printf("Present Days = %d\n", present);
214
215  printf("\n");
216
217  char course[20] = "C Programming";
218
219  printf("%c\n", course[20]);
220  // Error: Index 20 is out of bounds for an array of size 20.
221  // Solution: Access indices 0 to 19 only.
222
223  printf("\n");
224
```



```
225 char college[20];
226
227 college[0] = 'A';
228 college[1] = 'C';
229 college[2] = 'E';
230 college[3] = 'I';
231 college[4] = 'T';
232
233 printf("%s\n", college);
234 // Error: String is not null-terminated before printing with %.
235 // Solution: college[5] = '\0';
236
237 printf("\n");
238
239 int rollNo[5];
240
241 for(int i=0; i<5; i++)
242 {
243     scanf("%d", &rollNo[i]);
244 }
245
246 printf("\nRoll Numbers\n");
247
248 for(int i=0; i<5; i++)
249 {
250     printf("%d ", rollNo[i]);
251 }
252
```

```
253  printf("\n");
254
255  char section[3] = "IT-A";
256  // Error: Array size is too small for the string including null terminator.
257  // Solution: char section[5] = "IT-A";
258
259  printf("%s\n", section);
260
261  printf("\n");
262
263  printf("Program Finished\n");
264
265  return 0;
266 }
```